Sample Student Question & Answer Paper

Subject: Computer Science

Student: Alex Johnson

Total Questions: 5

Question 1

What is the primary purpose of a database index?

Student Answer: A database index is used to make searching and retrieving records faster. It creates an additional data structure that helps the database find rows without scanning the entire table.

Question 2

Explain the difference between a process and a thread.

Student Answer: A process is an independent program in execution with its own memory space. A thread is a smaller unit of execution inside a process and shares memory with other threads of that process.

Question 3

What is an API?

Student Answer: An API, or Application Programming Interface, is a way for two software systems to communicate with each other using defined rules, requests, and responses.

Question 4

What is the time complexity of binary search, and why?

Student Answer: Binary search has O(log n) time complexity because each comparison eliminates approximately half of the remaining search space.

Question 5

What is machine learning?

Student Answer: Machine learning is a branch of artificial intelligence where computer systems learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every case.